
Software Golden Modeling
Python Float32 Reference winograd.py / direct_conv.py Standard Winograd algorithm
Cross-validation
INT16 Fixed-Point Model fixed_point.py Emulates bit-growth, shifts, saturation
Test Vector Generator generate_vectors.py Randomized matrices & stimuli
input.hex
weight.hex
expected_output.hex
Automated Hardware Simulation
Self-Checking SystemVerilog TB winograd_core_tb.sv File I/O, Handshake sequencing
exp_data
Bit-Exact Assertion Checker Compare y_tile vs expected_output Automated mismatch counter [PASSED] 0 Mismatches
Stimulus (d_tile, u_tile)
Device Under Test (DUT) winograd_top.sv Pipelined RTL Datapath + FSM
Hardware Output (y_tile)